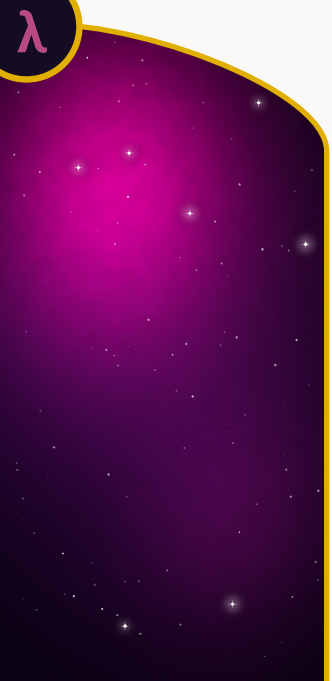


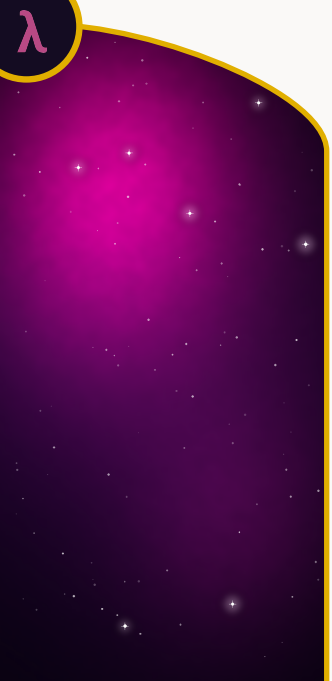
L11A: Streams II

CS1101S: Programming Methodology

Martin Henz

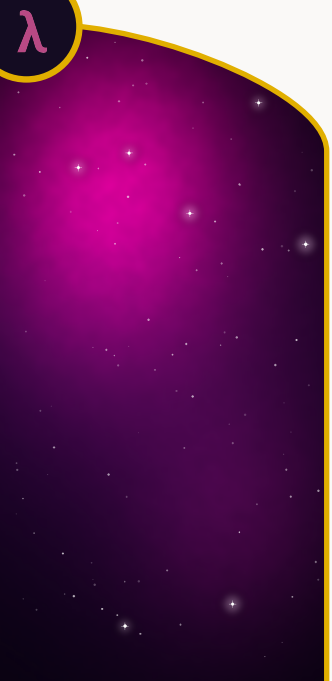
28th October, 2026

Version 1.0 — Built: 2026-10-25 15:52:45 SGT

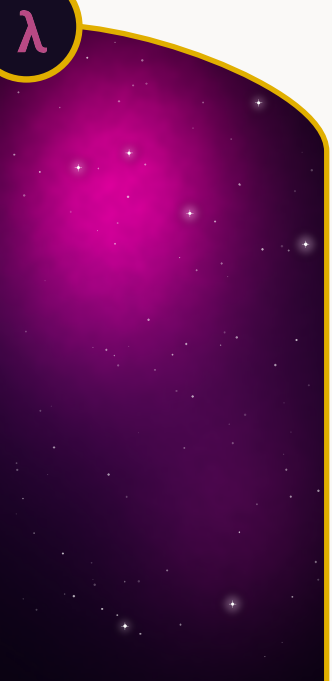


Announcements

- Mastery Check 2 begins now
- Topics
  - environment model
  - memoization
  - streams

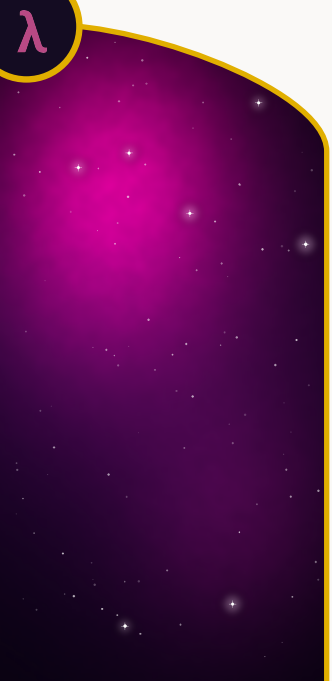


Course Overview



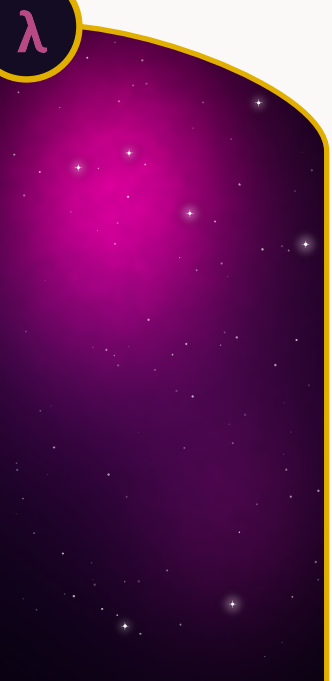
Course Overview

- Unit 1—Functions (textbook Chapter 1)



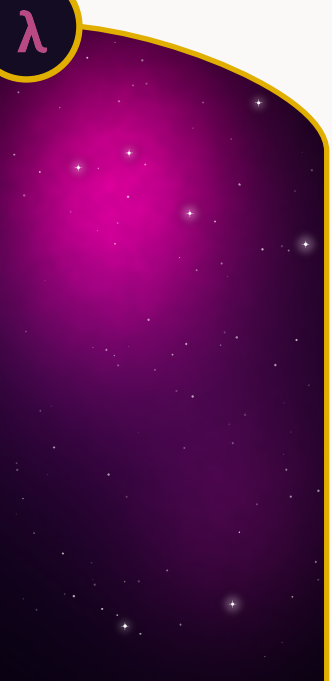
Course Overview

- Unit 1—Functions (textbook Chapter 1)
  - Getting acquainted with the elements of programming, using **functional abstraction**



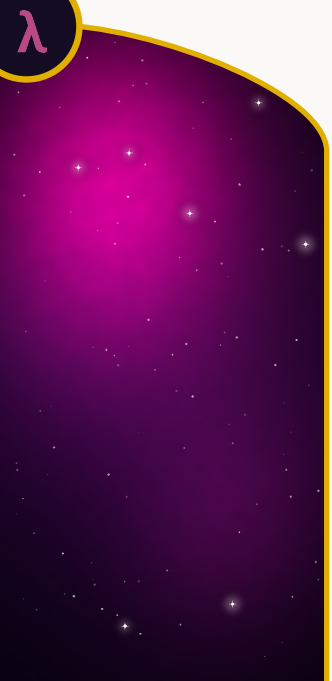
Course Overview

- Unit 1—Functions (textbook Chapter 1)
  - Getting acquainted with the elements of programming, using **functional abstraction**
  - Learning to read programs, and using the **substitution model**



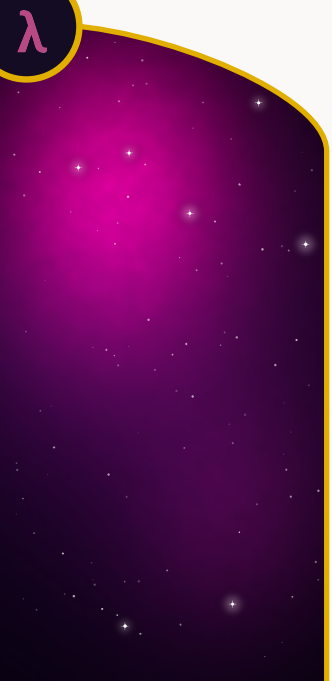
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  - Learning to read programs, and using the **substitution model**
  - Example applications: Runes, curves



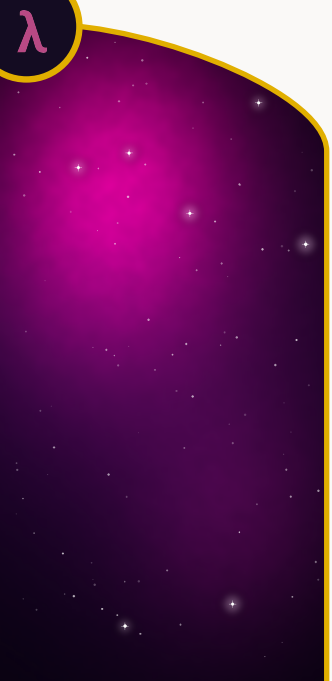
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- Unit 2—Data (textbook Chapter 2)



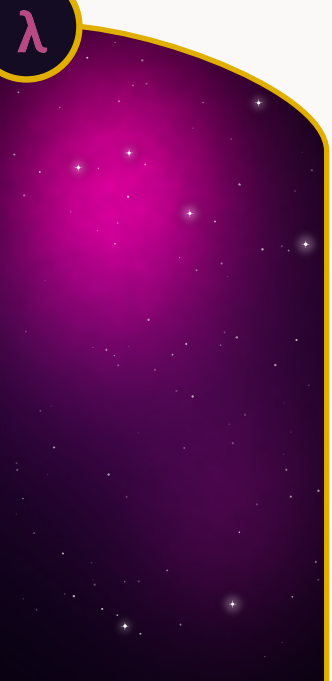
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- Unit 2—Data (textbook Chapter 2)
  - Getting familiar with data: pairs, linked lists, trees, streams



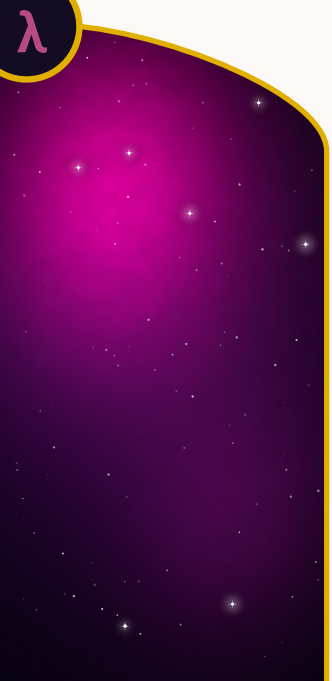
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  - Getting acquainted with the elements of programming, using **functional abstraction**
  - Learning to read programs, and using the **substitution model**
  - Example applications: Runes, curves
- Unit 2—Data (textbook Chapter 2)
  - Getting familiar with data: pairs, linked lists, trees, streams
  - Searching in linked lists and trees, sorting of linked lists

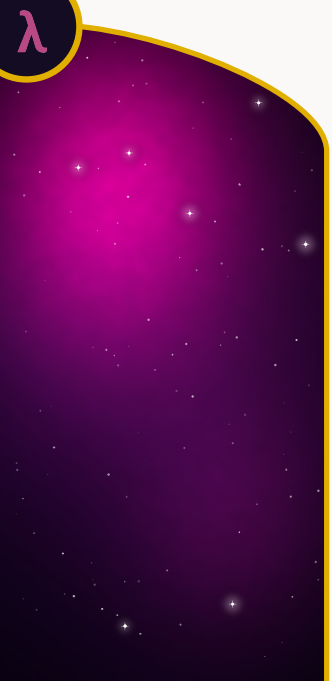


Course Overview

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  - Getting acquainted with the elements of programming, using **functional abstraction**
  - Learning to read programs, and using the **substitution model**
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  - Getting familiar with data: pairs, linked lists, trees, streams
  - Searching in linked lists and trees, sorting of linked lists
  - Example application: sound processing

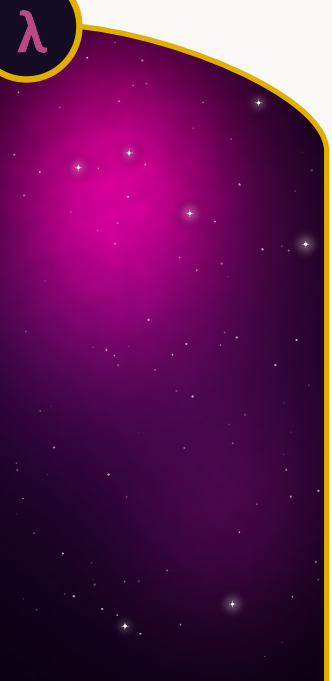


Course Overview, continued



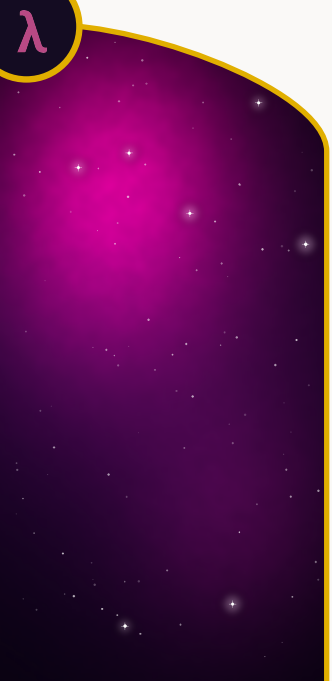
Course Overview, continued

- Unit 3—State (parts of textbook Chapter 3)



Course Overview, continued

- Unit 3—State (parts of textbook Chapter 3)
  - Programming with stateful abstractions



Course Overview, continued

- Unit 3—State (parts of textbook Chapter 3)
  - Programming with stateful abstractions
  - Lists, loops, searching in and sorting of lists

Course Overview, continued

- Unit 3—State (parts of textbook Chapter 3)
  - Programming with stateful abstractions
  - Lists, loops, searching in and sorting of lists
  - Reading programs using the environment model

Course Overview, continued

- Unit 3—State (parts of textbook Chapter 3)
  - Programming with stateful abstractions
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  - Example applications: robotics, video processing

Course Overview, continued

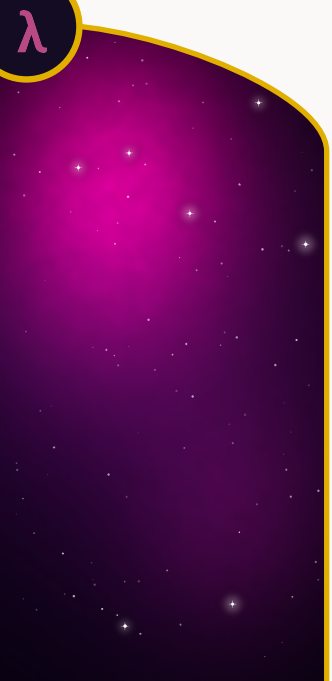
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- **Unit 4—Beyond (parts of textbook Chapters 3 and 4)**

Course Overview, continued

- Unit 3—State (parts of textbook Chapter 3)
  - Programming with stateful abstractions
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  - Example applications: robotics, video processing
- **Unit 4—Beyond (parts of textbook Chapters 3 and 4)**
  - Memoization, **Memoized Streams**

Course Overview, continued

- Unit 3—State (parts of textbook Chapter 3)
  - Programming with stateful abstractions
  - Lists, loops, searching in and sorting of lists
  - Reading programs using the environment model
  - Example applications: robotics, video processing
- **Unit 4—Beyond (parts of textbook Chapters 3 and 4)**
  - Memoization, **Memoized Streams**
  - Metalinguistic abstraction



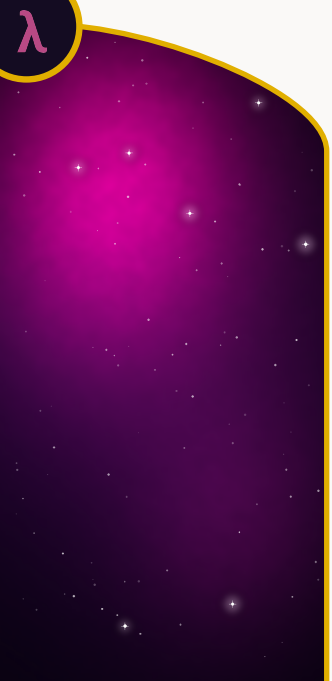
Outline

Review

Streams as Recursive Definitions

Memoization

Applications



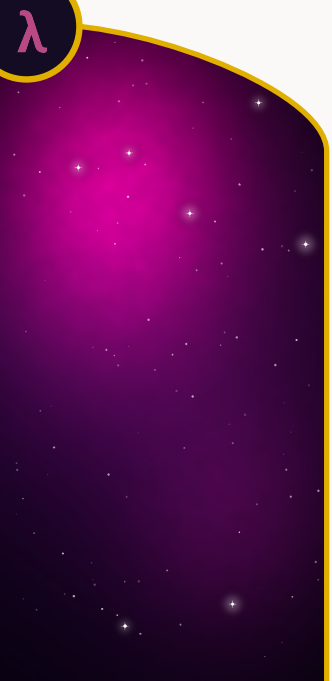
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Review

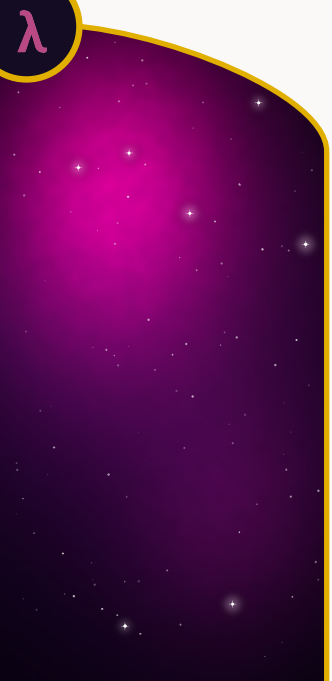
Streams as Recursive Definitions

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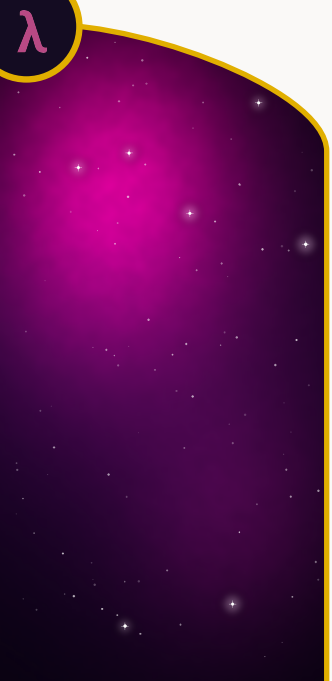


Idea of Streams



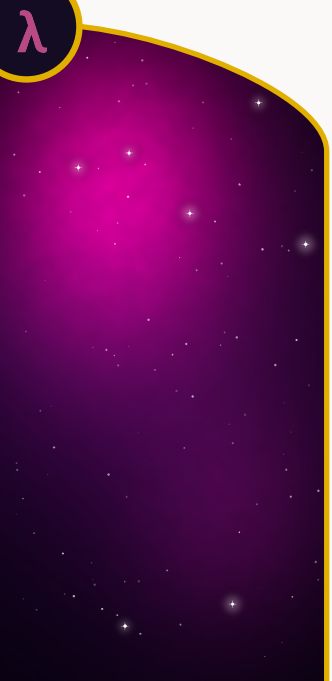
Idea of Streams

- Delayed linked lists



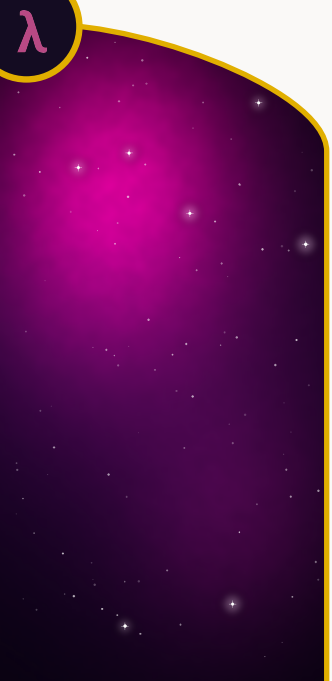
Idea of Streams

- Delayed linked lists
  - Our pairs contain a data item as `head` (as usual), but a function as `tail` that can be activated when needed



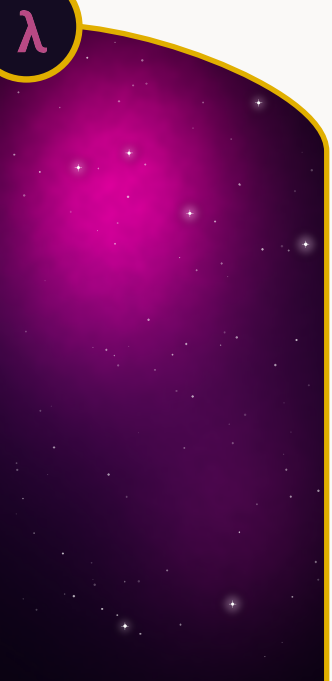
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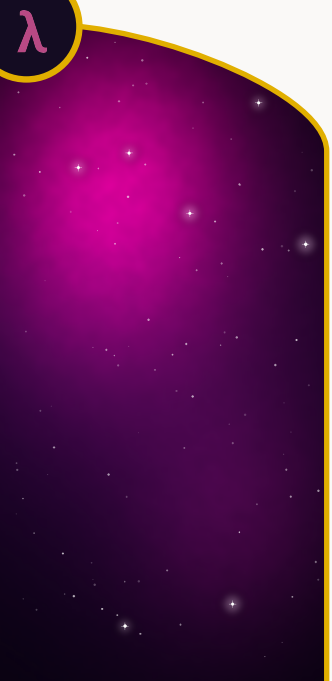
Idea of Streams

- Delayed linked lists
  - Our pairs contain a data item as `head` (as usual), but a function as `tail` that can be activated when needed
- Streams
  - A stream of a certain type is either `None`,



Idea of Streams

- Delayed linked lists
  - Our pairs contain a data item as `head` (as usual), but a function as `tail` that can be activated when needed
- Streams
  - A stream of a certain type is either `None`,
  - or a pair

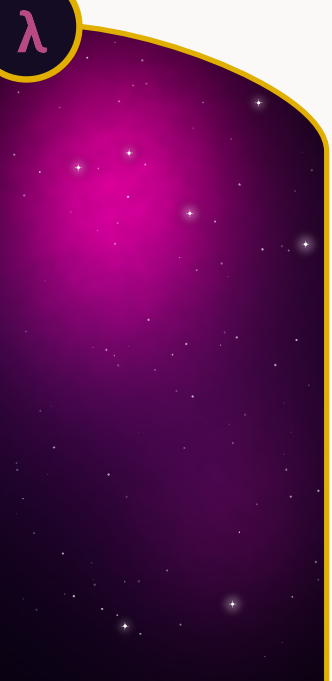


Idea of Streams

- Delayed linked lists
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- Streams
  - A stream of a certain type is either `None`,
  - or a pair
    - whose `head` is of that type and

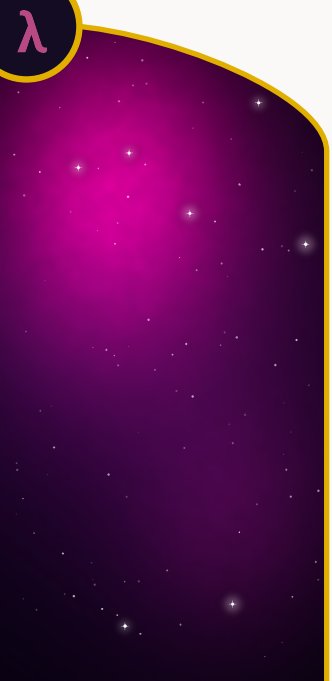
Idea of Streams

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  - Our pairs contain a data item as `head` (as usual), but a function as `tail` that can be activated when needed
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  - A stream of a certain type is either `None`,
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    - whose `tail` is a nullary function that returns a stream of that type



Idea of Streams

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- Stream discipline



Idea of Streams

- Delayed linked lists
  - Our pairs contain a data item as `head` (as usual), but a function as `tail` that can be activated when needed
- Streams
  - A stream of a certain type is either `None`,
  - or a pair
    - whose `head` is of that type and
    - whose `tail` is a nullary function that returns a stream of that type
- Stream discipline
  - Like the linked-list discipline, now using streams

Examples — Finite Streams

```
# An empty stream
s1 = None
```



Examples — Finite Streams

```
# An empty stream
s1 = None
```



```
# A stream with element 1
s2 = pair(1, lambda: None)
```



Examples — Finite Streams

```
# An empty stream
s1 = None
```

```
# A stream with element 1
s2 = pair(1, lambda: None)
```

```
# A stream with elements 1, 2, 3
s3 = pair(1,
          lambda: pair(2,
                       lambda: pair(3,
                                    lambda: None)))
```

Example — An Infinite Stream

```
def ones_stream():
    return pair(1, ones_stream)
ones = ones_stream()
```



Example — An Infinite Stream

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- head(ones) $\Rightarrow$ 1

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- head(ones) $\Rightarrow$ 1
- head(tail(ones)()) $\Rightarrow$ 1

Example — An Infinite Stream

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def ones_stream():
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```



- head(ones) $\Rightarrow$ 1
- head(tail(ones)()) $\Rightarrow$ 1
- head(tail(tail(ones)())()) $\Rightarrow$ 1

stream_tail

```
ones = ones_stream()
```



stream_tail

```
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```



- head(ones) $\Rightarrow$ 1

stream_tail

```
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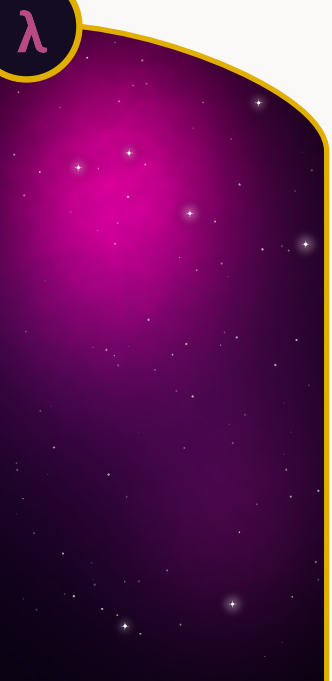
- head(ones) $\Rightarrow$ 1
- head(stream_tail(ones)) $\Rightarrow$ 1

stream_tail

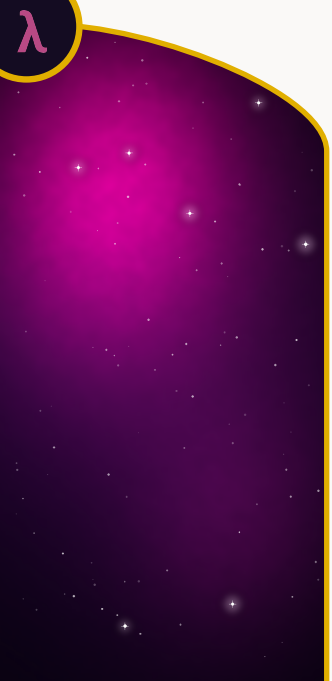
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- head(ones) $\Rightarrow$ 1
- head(stream_tail(ones)) $\Rightarrow$ 1
- head(stream_tail(stream_tail(ones))) $\Rightarrow$ 1

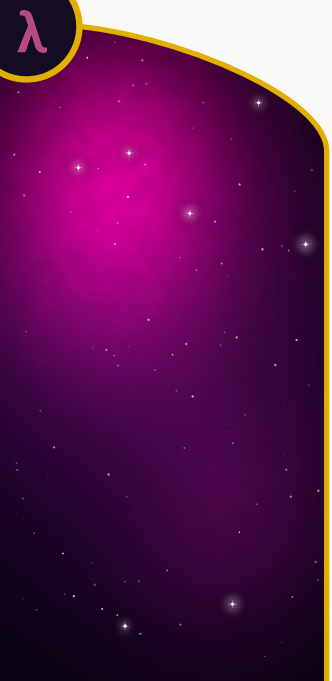


Stream Processing



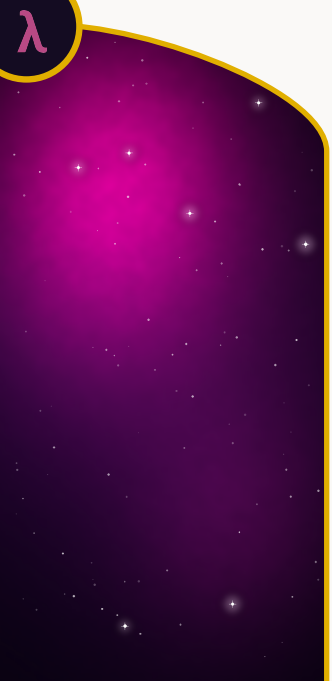
Stream Processing

- Like linked lists, except



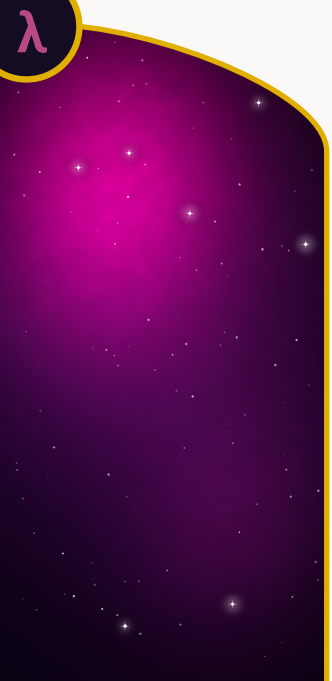
Stream Processing

- Like linked lists, except
  - Wrap `tail` in a function



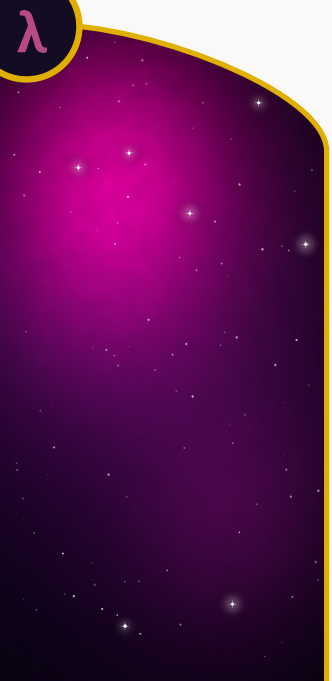
Stream Processing

- Like linked lists, except
  - Wrap `tail` in a function
  - Use `stream_tail` instead of `tail`



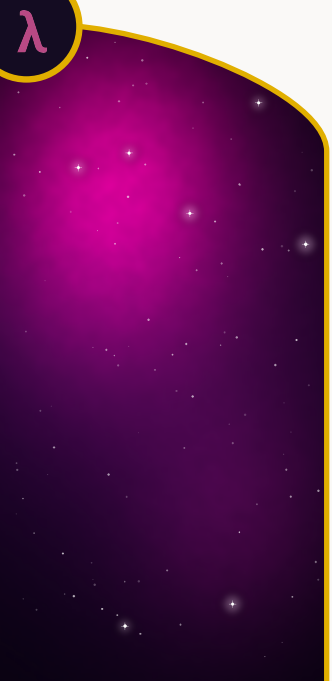
Stream Processing

- Like linked lists, except
  - Wrap `tail` in a function
  - Use `stream_tail` instead of `tail`
- Stream support in Python §3



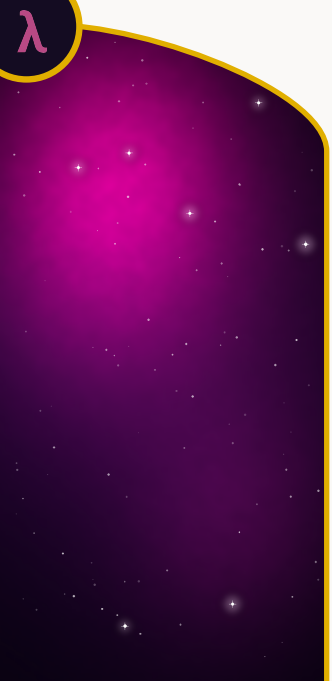
Stream Processing

- Like linked lists, except
  - Wrap `tail` in a function
  - Use `stream_tail` instead of `tail`
- Stream support in Python §3
  - Pre-declared functions for stream processing



Stream Processing

- Like linked lists, except
  - Wrap `tail` in a function
  - Use `stream_tail` instead of `tail`
- Stream support in Python §3
  - Pre-declared functions for stream processing
    - Examples: `stream_tail`, `stream_map`, `stream_filter`



Stream Processing

- Like linked lists, except
  - Wrap `tail` in a function
  - Use `stream_tail` instead of `tail`
- Stream support in Python §3
  - Pre-declared functions for stream processing
    - Examples: `stream_tail`, `stream_map`, `stream_filter`
  - Refer to Specification of Python §3 (`python_3.pdf`)

enum_stream

```
def enum_stream(low, hi):
    return None if low > hi else pair(low,
        lambda: enum_stream(low + 1, hi))
s = enum_stream(1, 100)
```



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- head(s) $\Rightarrow$ 1
- head(stream_tail(s)) $\Rightarrow$ 2

enum_stream

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def enum_stream(low, hi):
    return None if low > hi else pair(low,
        lambda: enum_stream(low + 1, hi))
s = enum_stream(1, 100)
```



- head(s) $\Rightarrow$ 1
- head(stream_tail(s)) $\Rightarrow$ 2
- head(stream_tail(stream_tail(s))) $\Rightarrow$ 3

stream_ref

```
def enum_stream(low, hi):
    return None if low > hi else pair(low,
        lambda: enum_stream(low + 1, hi))
def stream_ref(s, n):
    return head(s) if n == 0 else stream_ref(
        stream_tail(s), n - 1)
s = enum_stream(1, 100)
```



stream_ref

```
def enum_stream(low, hi):
    return None if low > hi else pair(low,
        lambda: enum_stream(low + 1, hi))
def stream_ref(s, n):
    return head(s) if n == 0 else stream_ref(
        stream_tail(s), n - 1)
s = enum_stream(1, 100)
```



- stream_ref(s, 0) ⇒ 1

stream_ref

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def enum_stream(low, hi):
    return None if low > hi else pair(low,
        lambda: enum_stream(low + 1, hi))
def stream_ref(s, n):
    return head(s) if n == 0 else stream_ref(
        stream_tail(s), n - 1)
s = enum_stream(1, 100)
```

- stream_ref(s, 0) ⇒ 1
- stream_ref(s, 10) ⇒ 11

stream_ref

```
def enum_stream(low, hi):
    return None if low > hi else pair(low,
        lambda: enum_stream(low + 1, hi))
def stream_ref(s, n):
    return head(s) if n == 0 else stream_ref(
        stream_tail(s), n - 1)
s = enum_stream(1, 100)
```

- stream_ref(s, 0) ⇒ 1
- stream_ref(s, 10) ⇒ 11
- stream_ref(s, 99) ⇒ 100

stream_map and stream_filter

```
def stream_map(f, s):
    if is_none(s):
        return None
    else:
        return pair(f(head(s)),
                    lambda: stream_map(f, stream_tail(s)))

def stream_filter(p, s):
    if is_none(s):
        return None
    elif p(head(s)):
        return pair(head(s),
                    lambda: stream_filter(p,
    stream_tail(s)))
    else:
        return stream_filter(p, stream_tail(s))
```

More Examples

```
def integers_from(n):
    return pair(n, lambda: integers_from(n + 1))
integers = integers_from(1)
```



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- `stream_ref(integers, 0)` $\Rightarrow$ 1

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- `stream_ref(integers, 0)` $\Rightarrow$ 1
- `stream_ref(integers, 10)` $\Rightarrow$ 11

More Examples

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    return pair(n, lambda: integers_from(n + 1))
integers = integers_from(1)
```



- `stream_ref(integers, 0)` $\Rightarrow$ 1
- `stream_ref(integers, 10)` $\Rightarrow$ 11
- `stream_ref(integers, 99)` $\Rightarrow$ 100

More Examples

```
def is_divisible(x, y):
    return x % y == 0
no_fours = stream_filter(lambda x: not is_divisible(x, 4),
                         integers)
```



More Examples

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def is_divisible(x, y):
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- `stream_ref(no_fours, 3)` $\Rightarrow 5$

More Examples

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def is_divisible(x, y):
    return x % y == 0
no_fours = stream_filter(lambda x: not is_divisible(x, 4),
                         integers)
```



- stream_ref(no_fours, 3) $\Rightarrow$ 5
- stream_ref(no_fours, 100) $\Rightarrow$ 134

From Streams to Linked Lists

```
def eval_stream(s, n):
    return None if n == 0 else pair(head(s),
        eval_stream(stream_tail(s), n - 1))
```

```
eval_stream(no_fours, 10)
```

From Streams to Linked Lists

```
def eval_stream(s, n):
    return None if n == 0 else pair(head(s),
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```
eval_stream(no_fours, 10)
```

```
[1, [2, [3, [5, [6, [7, [9, [10, [11, [13,
None]]]]]]]]]]
```

Fibonacci Numbers

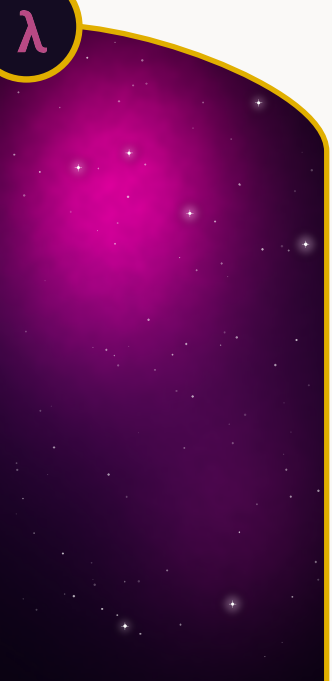


```
def fibgen(a, b):
    return pair(a, lambda: fibgen(b, a + b))
fibs = fibgen(0, 1)
eval_stream(fibs, 10)
```

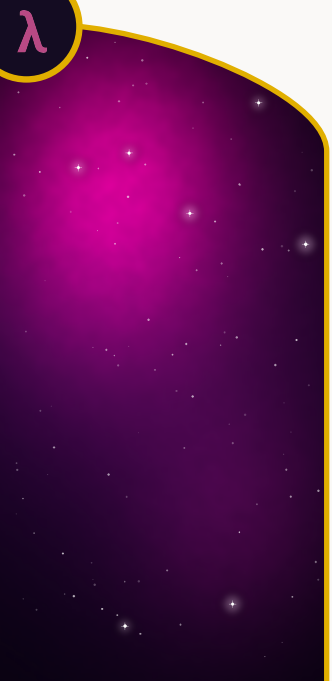
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More and More...



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- Wanted: Stream containing 1, 1, 2, 1, 2, 3, 1, 2, 3, 4, ...

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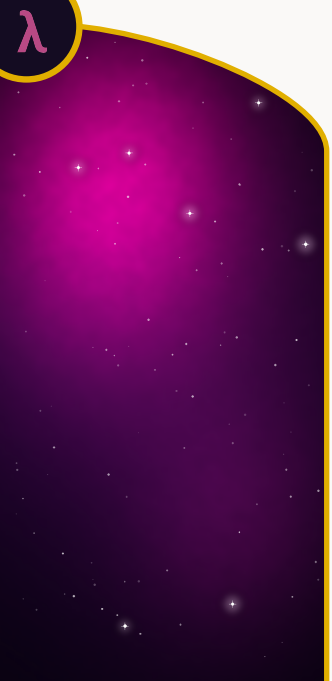
```
def more(a, b):
    return more(1, 1 + b) if a > b else pair(a,
        lambda: more(a + 1, b))
more_and_more = more(1, 1)
eval_stream(more_and_more, 15)
```

More and More...

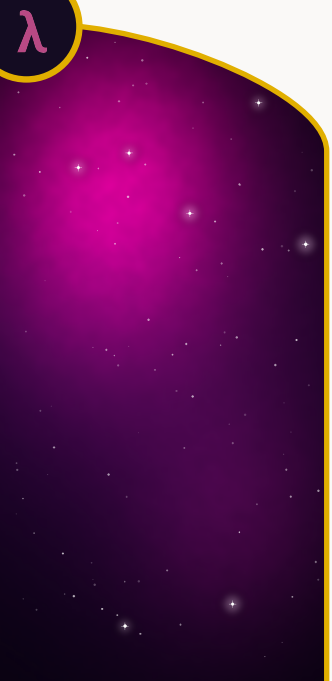
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```

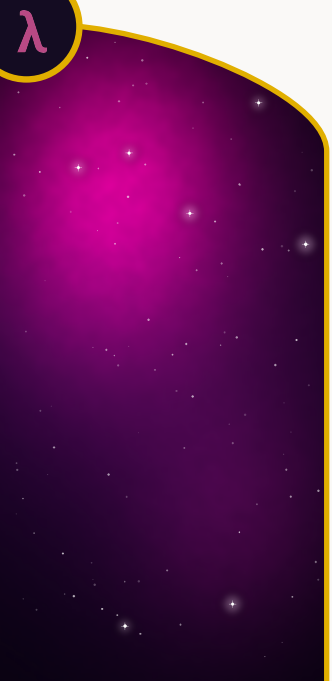


Replace



Replace

- Wanted: A function `replace` that creates a new stream by replacing in a given stream a particular value by another value



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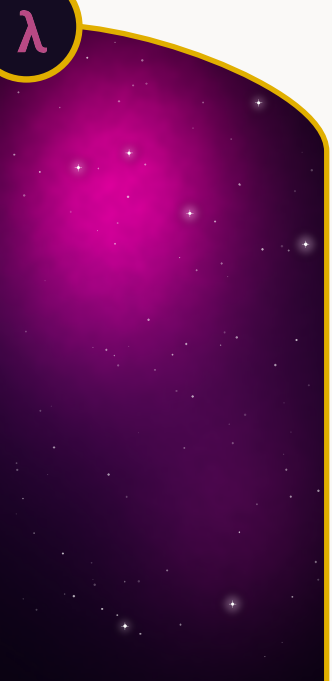
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- Example:

```
replace(more_and_more, 1, 0)
```



- produces the stream that contains
- 0, 0, 2, 0, 2, 3, 0, 2, 3, 4, 0, 2, 3, 4, 5, ...



Replace

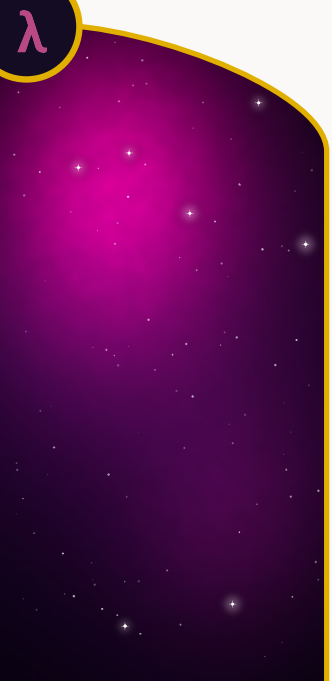
Replace

```
def replace(s, a, b):
    if is_none(s):
        return None
    else:
        return pair(b if head(s) == a else head(s),
                    lambda: replace(stream_tail(s), a, b))
s = replace(more_and_more, 1, 0)
eval_stream(s, 15)
```

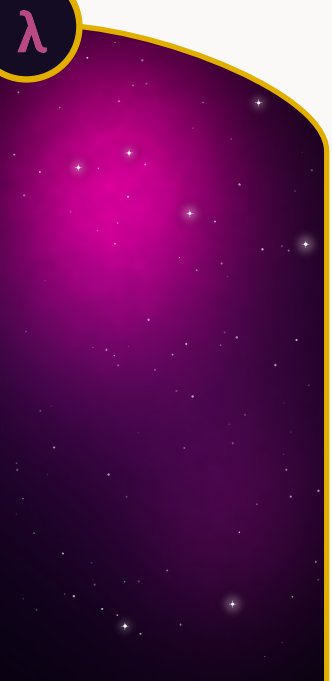
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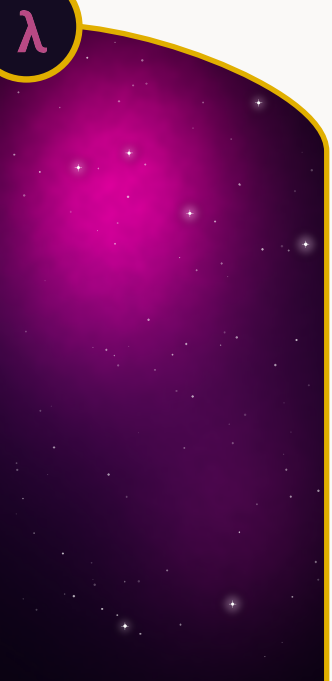


Linked List to Infinite Stream



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```
list_to_inf_stream(llist(1, 2, 3))
```



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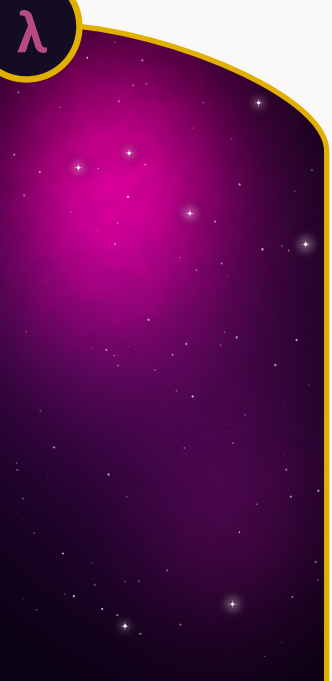
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```
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```

- produces the stream that contains
- 1, 2, 3, 1, 2, 3, 1, 2, 3, 1, 2, 3, 1, 2, 3, ...



Linked List to Infinite Stream

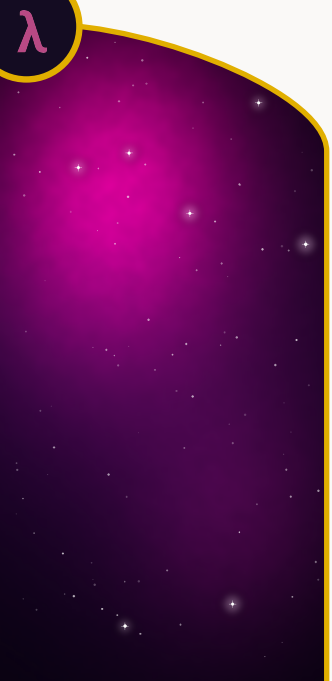
Linked List to Infinite Stream

```
def list_to_inf_stream(xs):
    def helper(ys):
        return helper(xs) if is_none(ys) else
    pair(head(ys),
             lambda: helper(tail(ys)))
    return None if is_none(xs) else helper(xs)
s = list_to_inf_stream(llist(1, 2, 3))
eval_stream(s, 10)
```

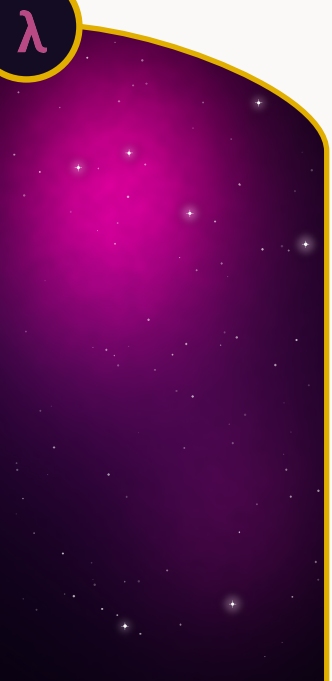
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s = list_to_inf_stream(llist(1, 2, 3))
eval_stream(s, 10)
```

```
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```



Why do we need streams?

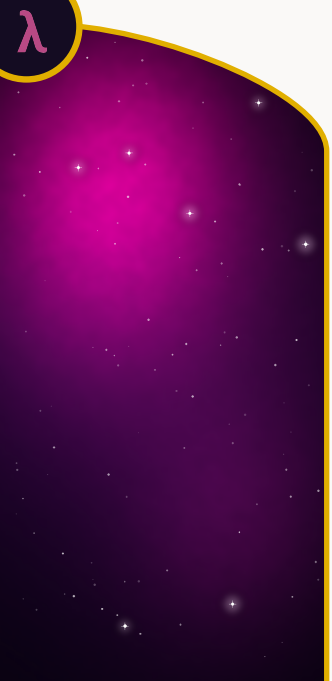


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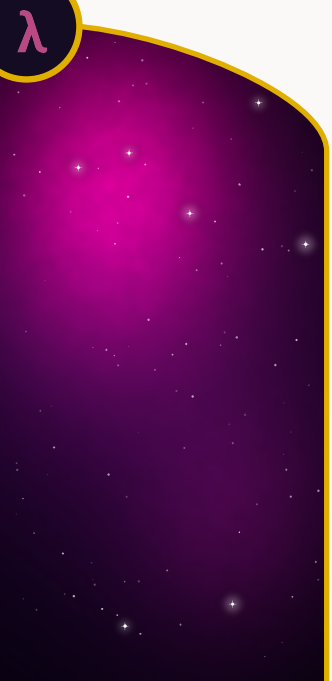


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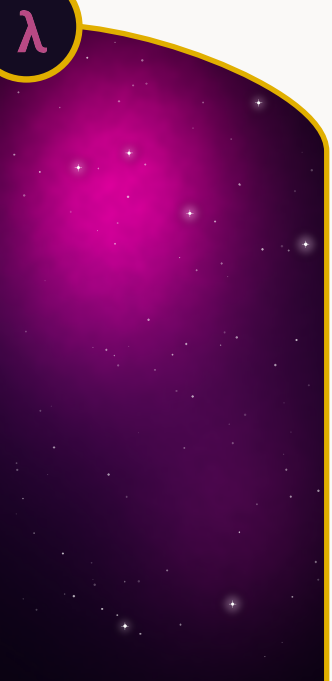
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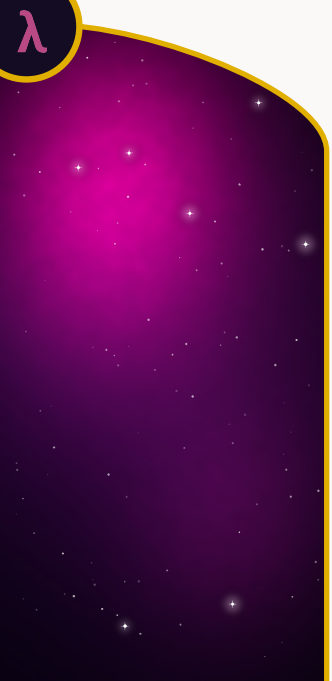
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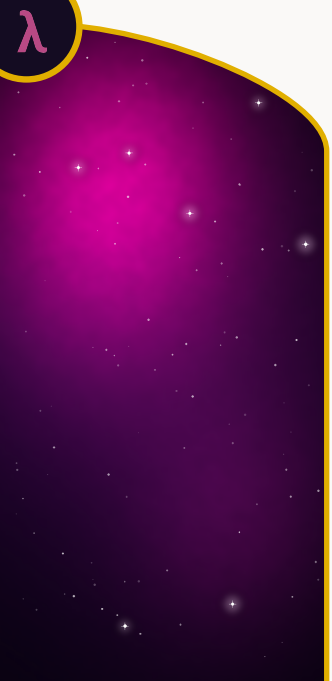
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- Example applications
- Video & Audio streaming
- Process and play A/V on-the-fly without entire file available
- Stream editing
- Process a file as a stream (e.g. find and replace text using AWK)



Outline

Review

Streams as Recursive Definitions

Memoization

Applications

Recall Stream of Ones



```
# original version
def ones_stream():
    return pair(1, ones_stream)
ones = ones_stream()
```

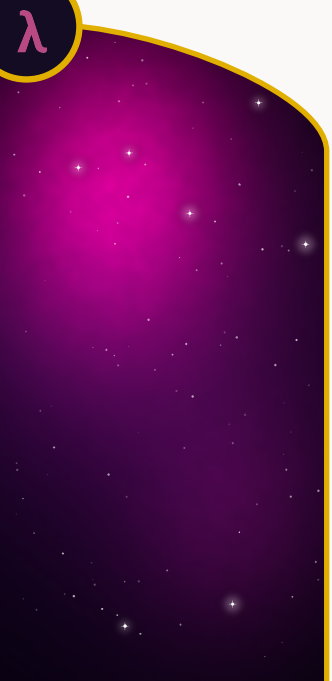
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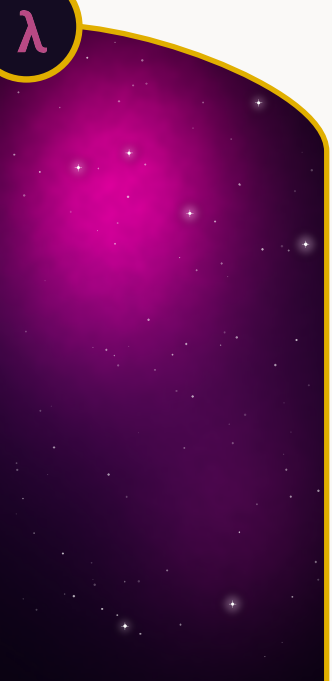


```
# alternative version
ones = pair(1, lambda: ones)
```





Defining a Repeating Sequence



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- Wanted: Stream containing 1, 2, 3, 1, 2, 3, 1, 2, 3, ...

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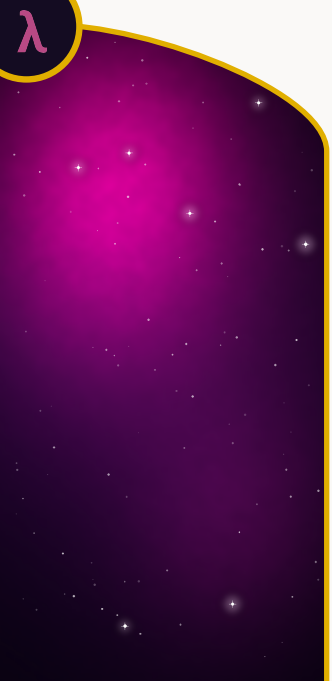
```
# A stream containing 1, 2, 3, 1, 2, 3, and so on
rep123 = pair(1, lambda: pair(2,
                        lambda: pair(3, lambda: rep123)))
eval_stream(rep123, 15)
```

Defining a Repeating Sequence

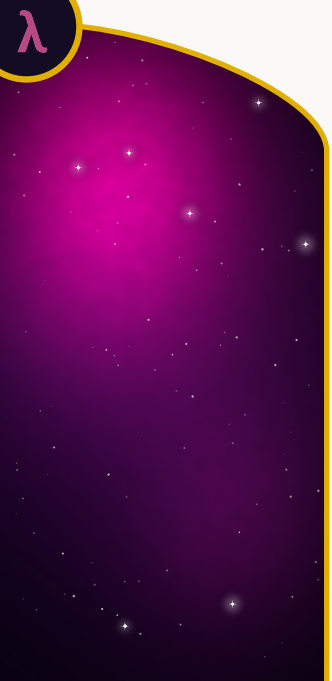
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```
# A stream containing 1, 2, 3, 1, 2, 3, and so on
rep123 = pair(1, lambda: pair(2,
                        lambda: pair(3, lambda: rep123)))
eval_stream(rep123, 15)
```

```
[1, [2, [3, [1, [2, [3, [1, [2, [3, [1, [2, [3, [1,
[2, [3, None]]]]]]]]]]]]]]
```

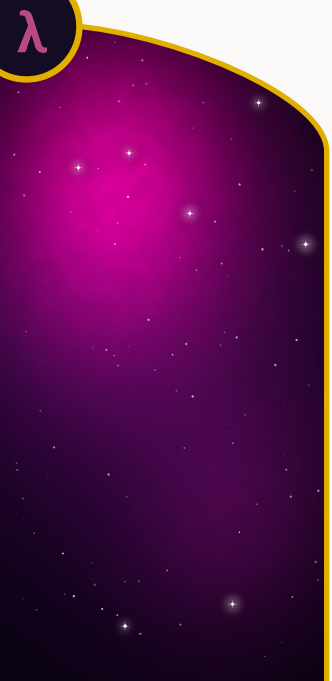


Adding Two Streams



Adding Two Streams

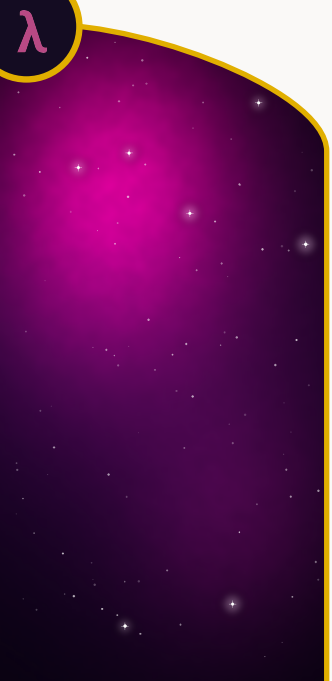
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- Example: Adding

```
1 2 3 4 5 6 ...
1 1 2 2 3 3 ...
```



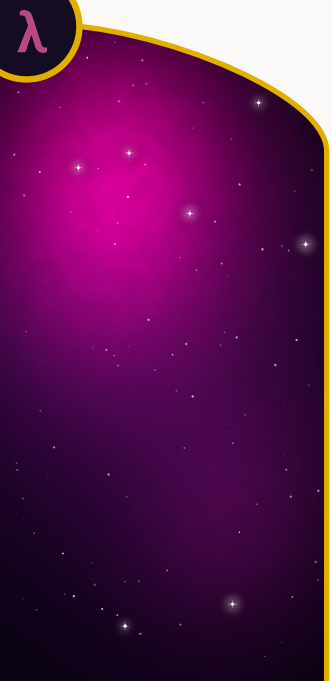
Adding Two Streams

- Wanted: A function `add_streams` that takes two streams and returns a stream that contains the element-wise sums
- Example: Adding

  ```
  1 2 3 4 5 6 ...
  1 1 2 2 3 3 ...
  ```

- should return

  ```
  2 3 5 6 8 9 ...
  ```



Adding Streams

Adding Streams

```
def add_streams(s1, s2):
    if is_none(s1):
        return s2
    elif is_none(s2):
        return s1
    else:
        return pair(head(s1) + head(s2),
                    lambda: add_streams(stream_tail(s1),
                                        stream_tail(s2)))
```

Fibonacci Numbers, Again

```
fibs = pair(0, lambda: pair(1,
    lambda: add_streams(stream_tail(fibs), fibs)))
eval_stream(fibs, 10)
```



Fibonacci Numbers, Again

```
fibs = pair(0, lambda: pair(1,
    lambda: add_streams(stream_tail(fibs), fibs)))
eval_stream(fibs, 10)
```

```
[0, [1, [1, [2, [3, [5, [8, [13, [21, [34,
None]]]]]]]]]]
```

Fibonacci Numbers, Again

```
  1  1  2  3  5   8  13 ... = stream_tail(fibs)
+ 0  1  1  2  3   5   8 ... = fibs
-----------------------------------------------
  0  1  1  2  3  5  8  13  21 ... = fibs
```

```
fibs = pair(0, lambda: pair(1,
    lambda: add_streams(stream_tail(fibs), fibs)))
```



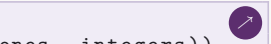
Integers Revisited

```
ones = pair(1, lambda: ones)
integers = pair(1, lambda: add_streams(ones, integers))
eval_stream(integers, 10)
```



Integers Revisited

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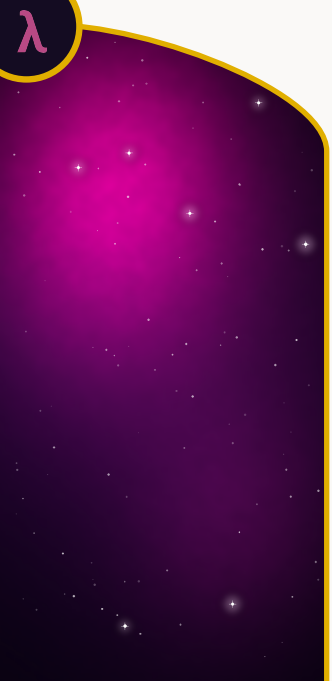
```
[1, [2, [3, [4, [5, [6, [7, [8, [9, [10,
None]]]]]]]]]]
```

Integers Revisited

```
    1  1  1  1  1  1 ... = ones
+   1  2  3  4  5  6 ... = integers
-------------------------------------
    1  2  3  4  5  6  7 ... = integers
```

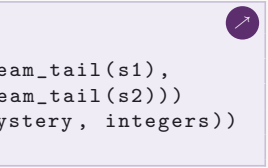
```
ones = pair(1, lambda: ones)
integers = pair(1, lambda: add_streams(ones, integers))
```





Which Stream Is This?

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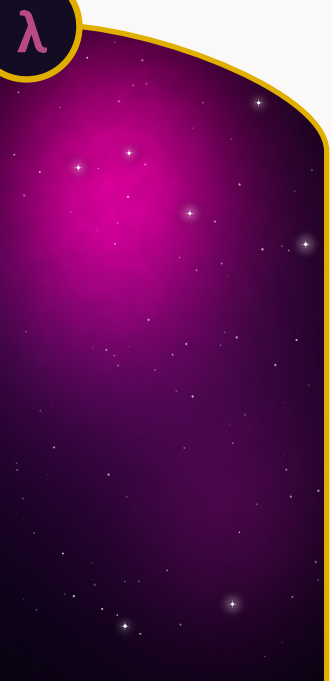
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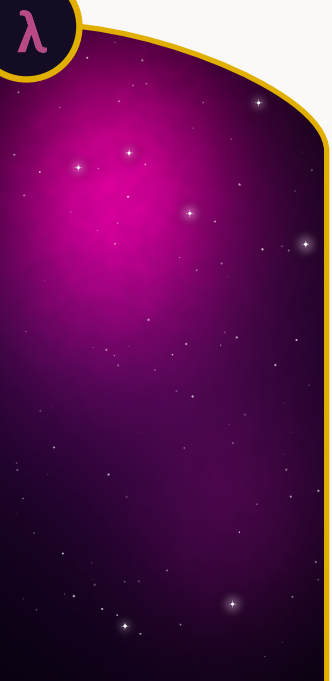
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Check-In time!

Factor in some fun!

Look for Check-In I11A in the Source Academy!



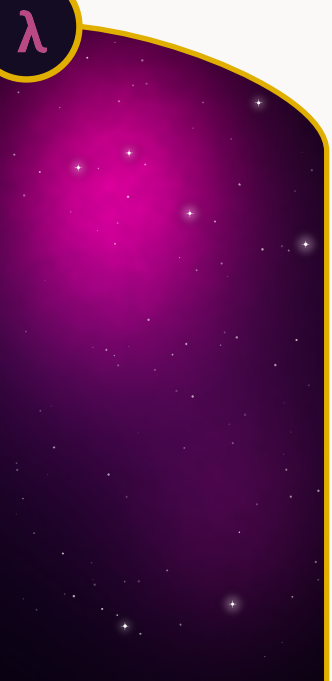
Outline

Review

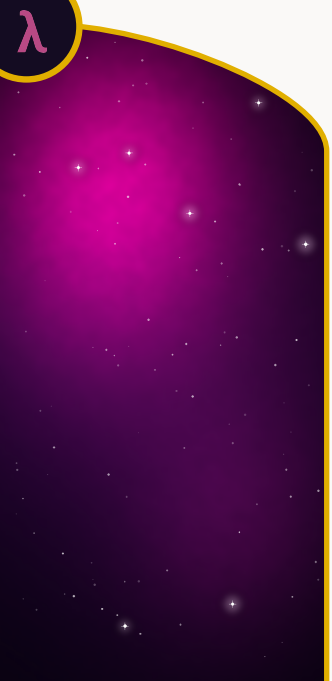
Streams as Recursive Definitions

Memoization

Applications

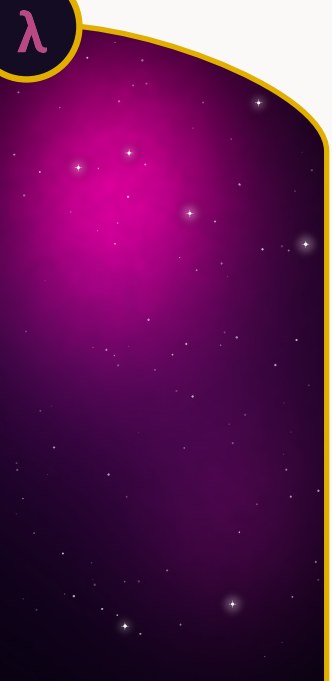


Recomputing Tails



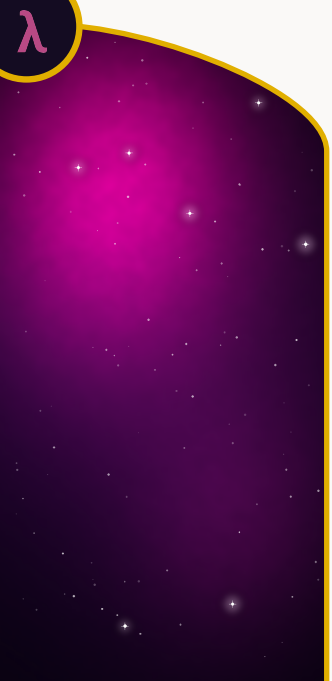
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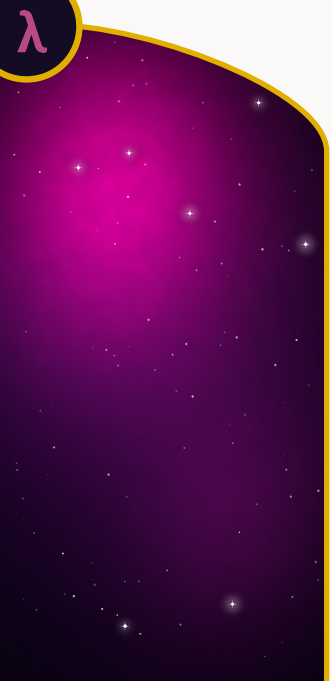
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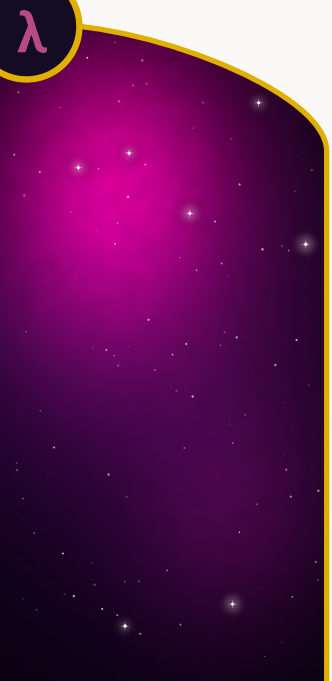


Recomputing Tails

- Every time we walk past an element of a stream, its tail function runs
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- Idea: use *memoization* to remember the result of the first call

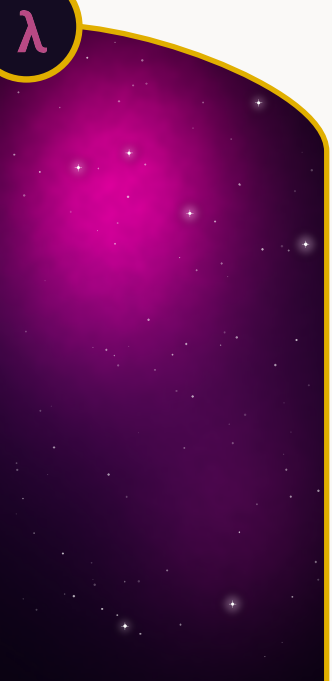


Without Memoization



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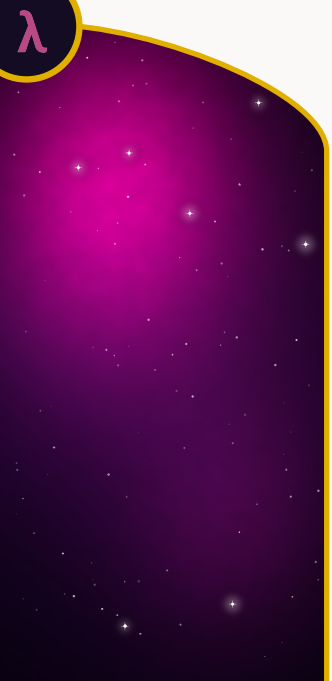
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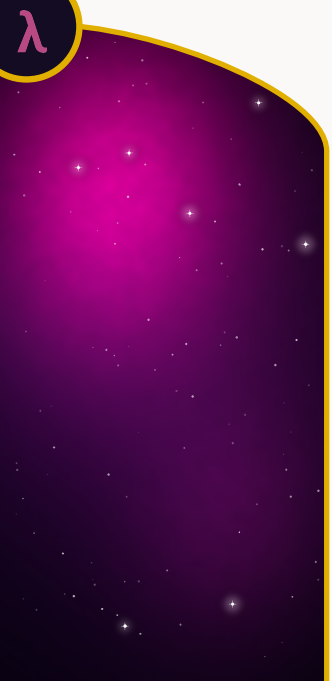
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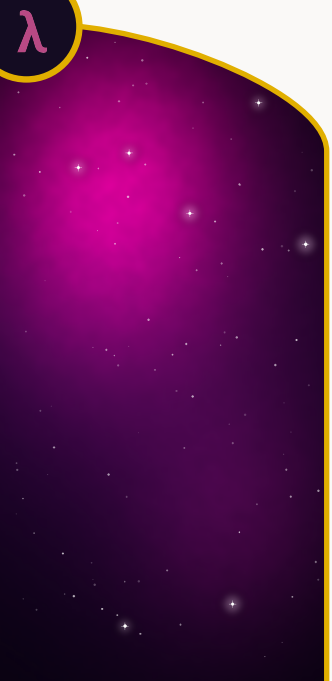


Memoization: `memo_fun`



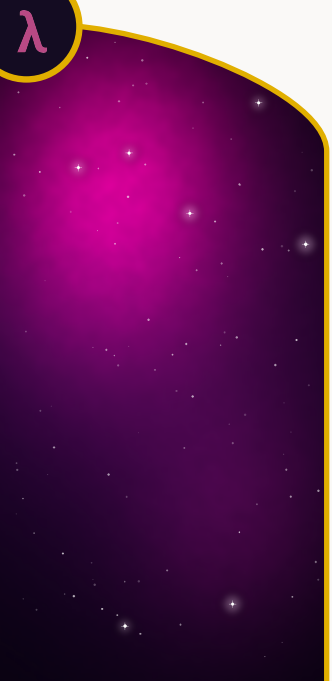
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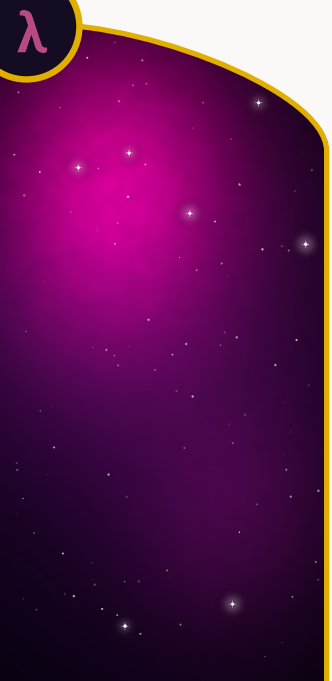
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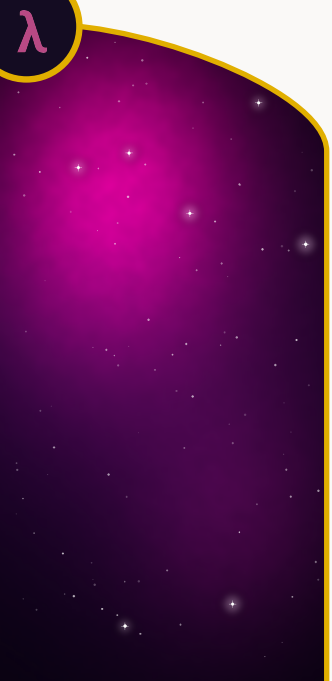
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def memo_fun(fun):
    already_run = False
    result = None
    def mfun():
        nonlocal already_run, result
        if already_run:
            return result
        else:
            result = fun()
            already_run = True
            return result
    return mfun
```





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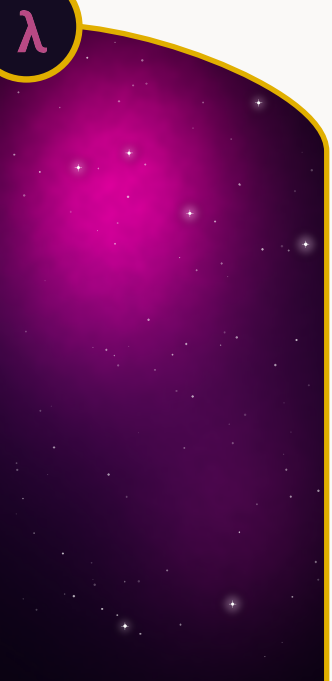
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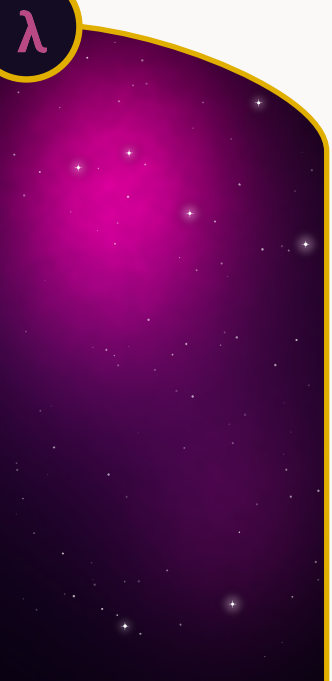
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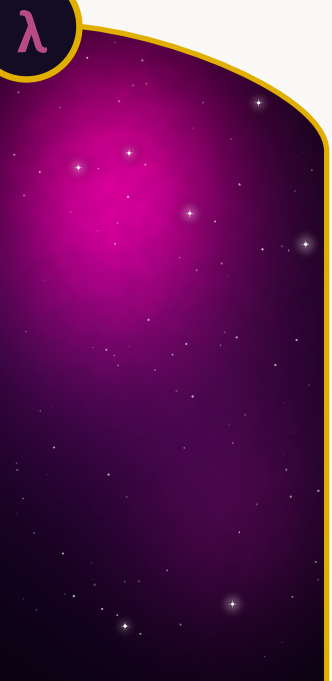


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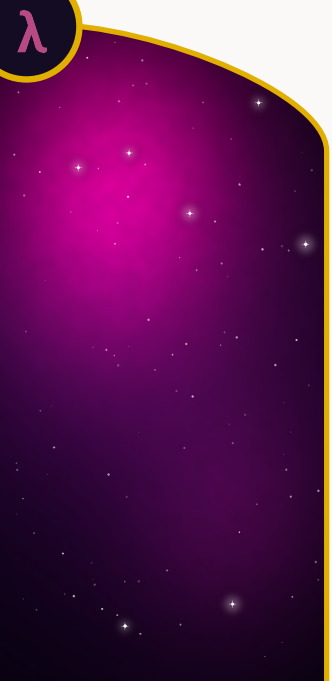
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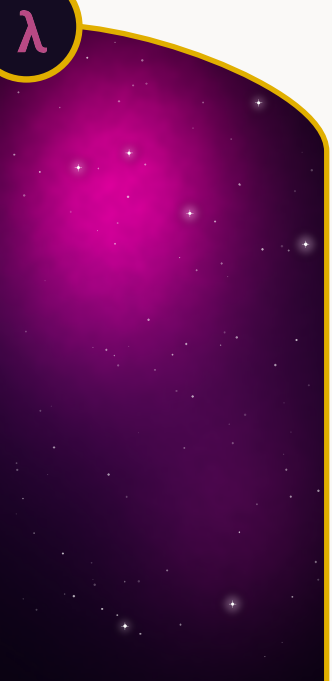
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- Pure functions have no side effects
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Outline

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Streams as Recursive Definitions

Memoization

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Example — Generating Primes

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def is_prime(n):
    def check(d):
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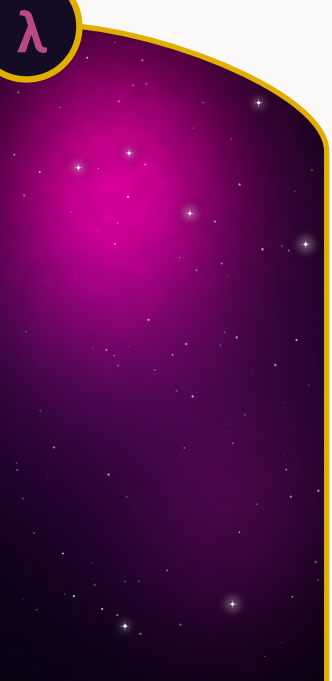
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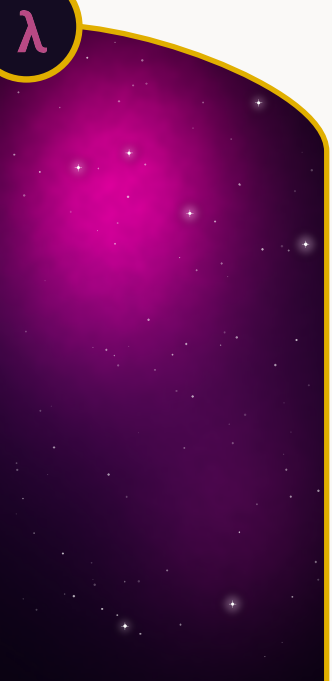
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Can we make a prime stream without using `is_prime`?



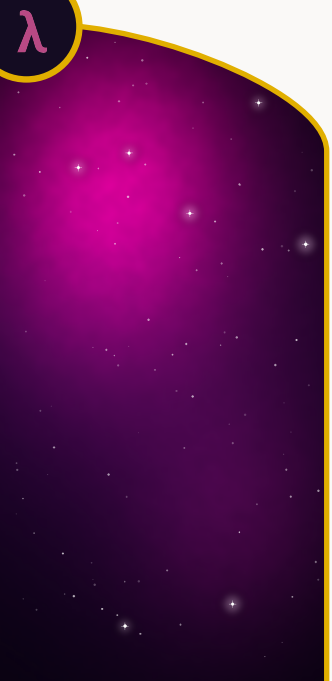
Sieving for Primes: Multiples of 2



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- Start with integers beginning with 2

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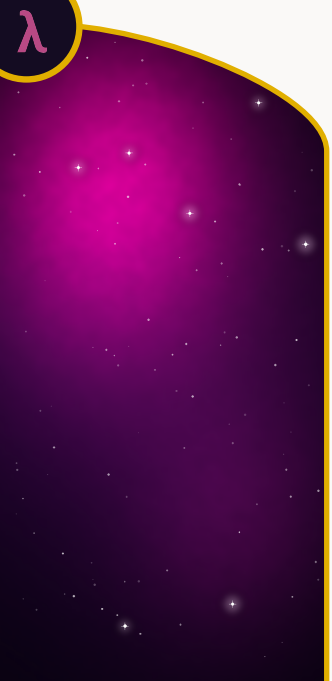
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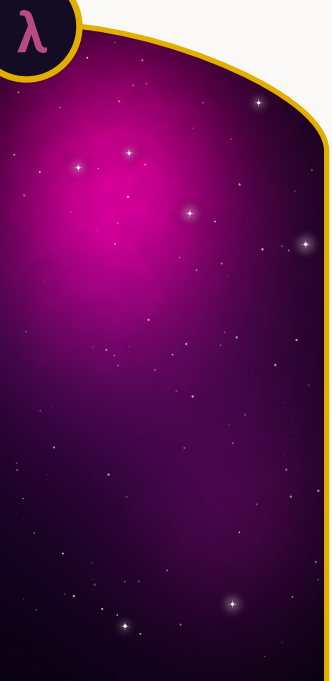
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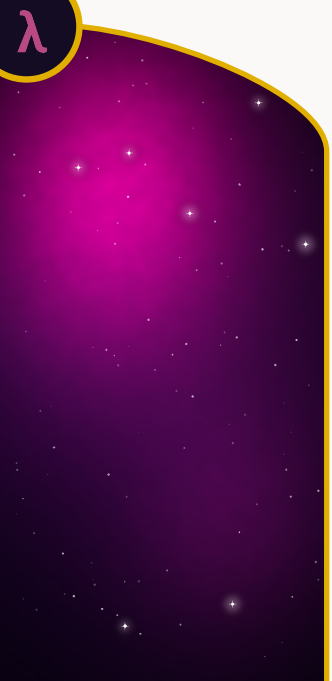
Sieving for Primes: Multiples of 3



Sieving for Primes: Multiples of 3

- Next left over integer is 3

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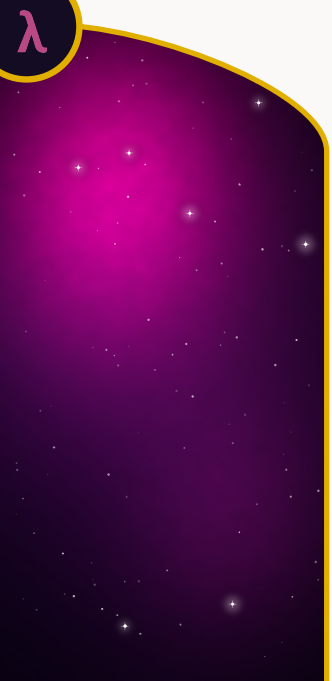
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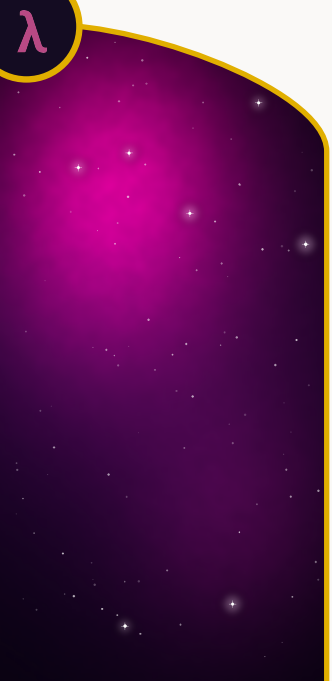
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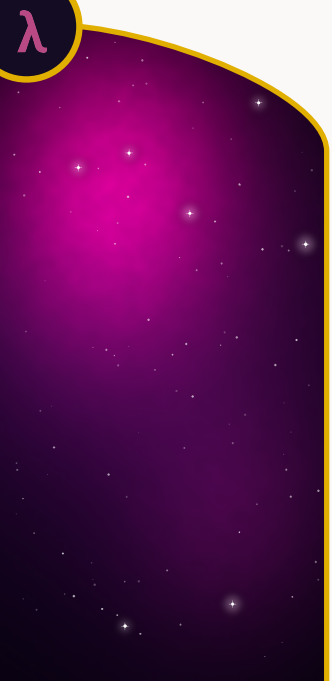
Sieving for Primes: Multiples of 5



Sieving for Primes: Multiples of 5

- Next left over integer is 5

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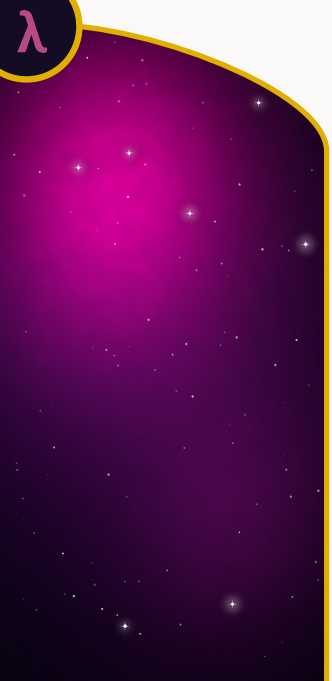


Sieving for Primes: Multiples of 5

- Next left over integer is 5

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- And so on...



Sieving for Primes

Sieving for Primes

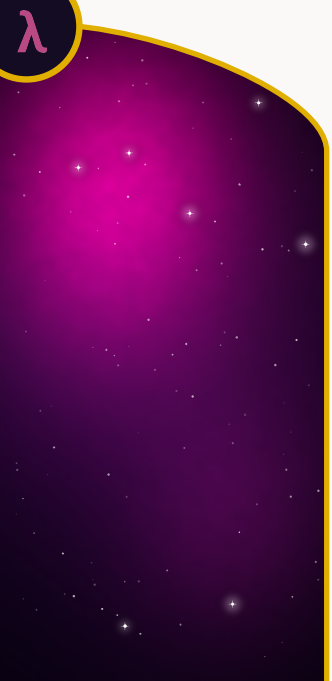


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def sieve(s):
    return pair(head(s),
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primes = sieve(integers_from(2))
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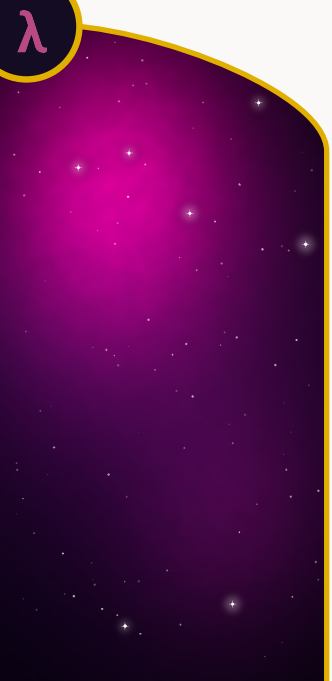
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```
def is_divisible(x, y):
    return x % y == 0
def sieve(s):
    return pair(head(s),
                lambda: sieve(stream_filter(
                    lambda x: not is_divisible(x,
    head(s)),
                    stream_tail(s))))
primes = sieve(integers_from(2))
eval_stream(primes, 10)
```

```
[2, [3, [5, [7, [11, [13, [17, [19, [23, [29,
None]]]]]]]]]]
```

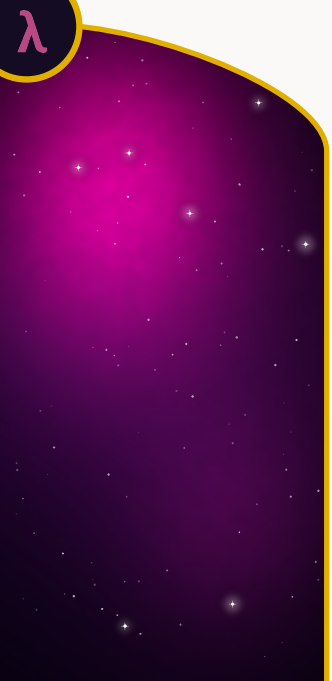


Is This the Sieve of Eratosthenes?



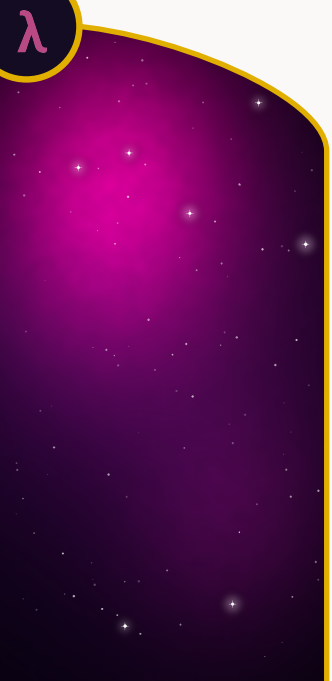
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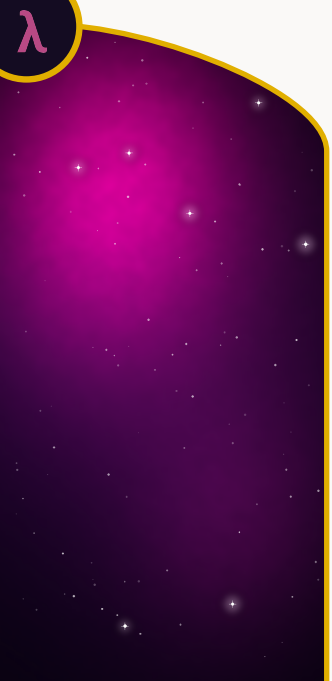
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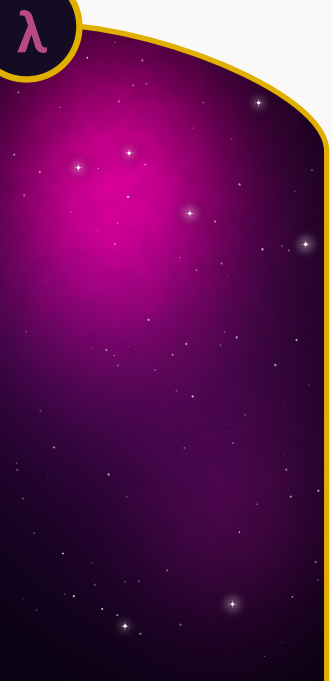
- Not actually the sieve of Eratosthenes
  - The sieve crosses off multiples of each prime
  - `sieve` tests each candidate for divisibility by the primes found so far: *trial division*, substantially less efficient (O'Neill, *The Genuine Sieve of Eratosthenes*, 2009)



Square Roots by Newton's Method

Square Roots by Newton's Method

```
def average(a, b):
    return (a + b) / 2
def good_enough(guess, x):
    return abs(guess * guess - x) < 0.001
def improve(guess, x):
    return average(guess, x / guess)
def sqrt_iter(guess, x):
    if good_enough(guess, x):
        return guess
    else:
        return sqrt_iter(improve(guess, x), x)
def sqrt(x):
    return sqrt_iter(1.0, x)
```



Using Streams for Iteration

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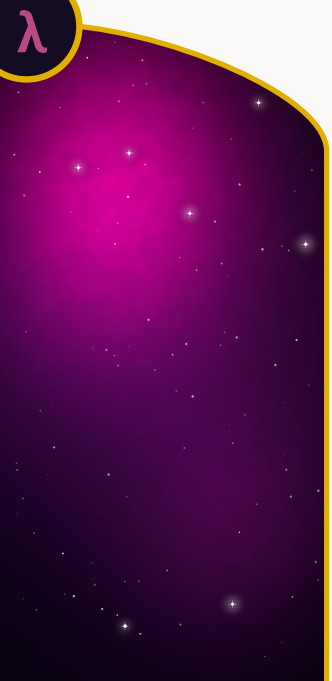
```
def sqrt_stream(x):
    guesses = pair(1.0, lambda: stream_map(
        lambda guess: improve(guess, x), guesses))
    return guesses
eval_stream(sqrt_stream(2), 6)
```



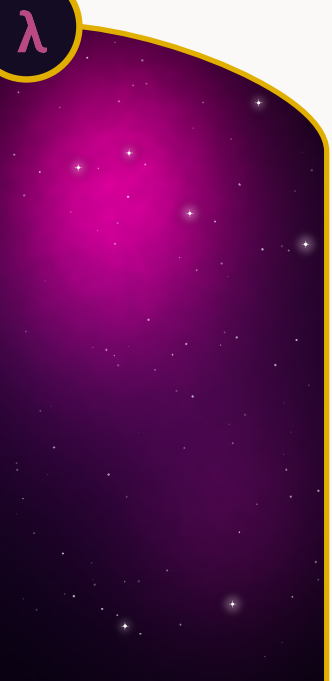
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```

```
[1.0, [1.5, [1.4166666666666665,
[1.4142156862745097, [1.4142135623746899,
[1.414213562373095, None]]]]]]
```

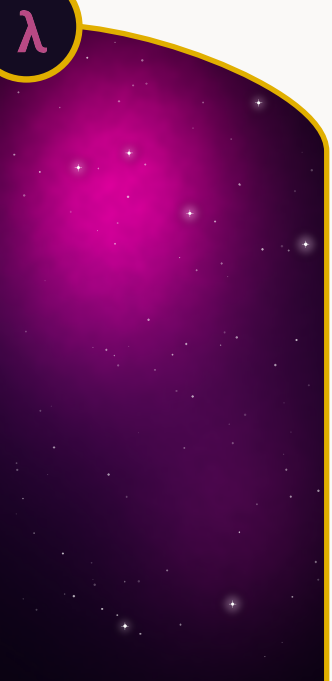


Summary



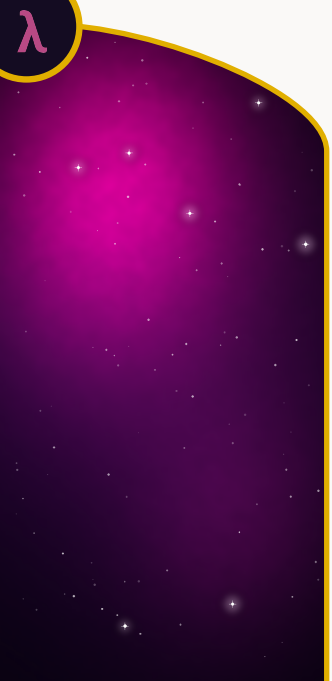
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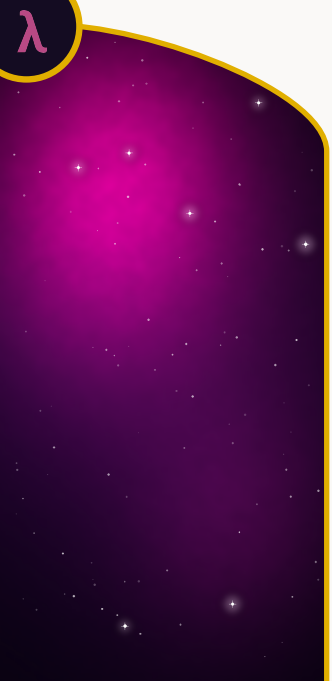
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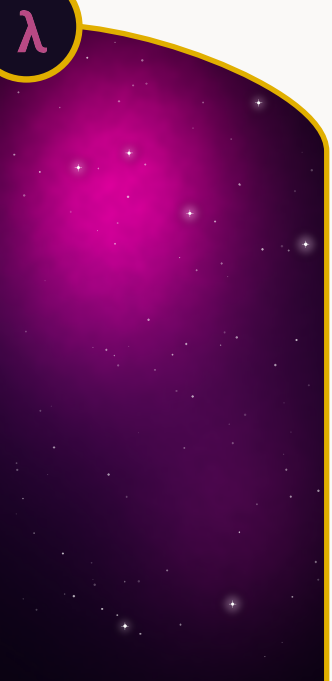
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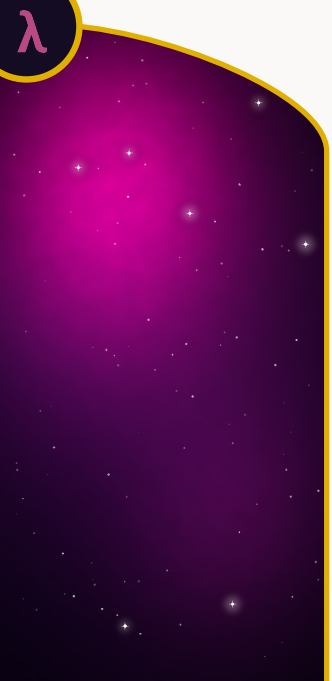
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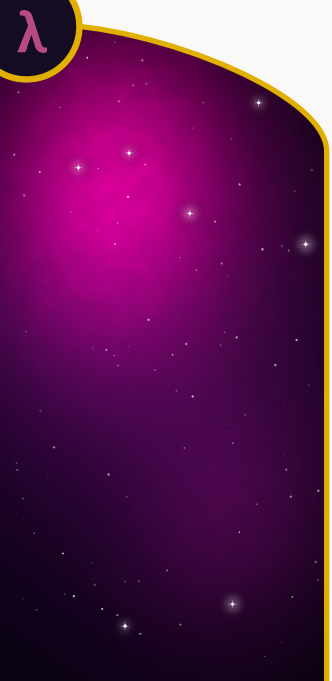
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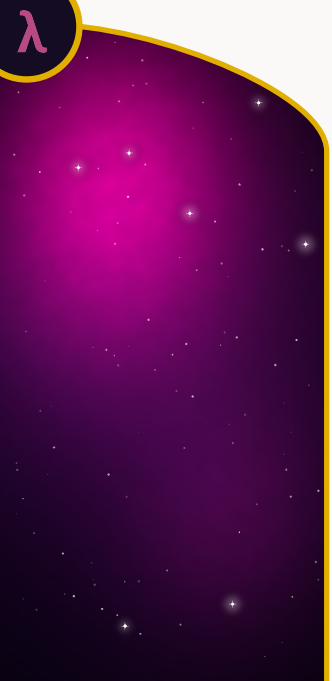
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- Memoization can be used on streams
  - Each tail function runs at most once
  - It changes behavior only if the tail has side effects
- Streams can be used for iteration: `sieve`, `sqrt_stream`